University Hospital Summary Report

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1. Introduction

In this module I am a data analyst researching risk factors for heart disease at a university hospital. I have access to a large set of historical data that can help analyze patterns between different health indicators and the presence of heart disease. I will be running two logistic regression models, a random forest classification model, and a random forest regression model. Results from these models could eventually be used to evaluate medical records and look for risks that might not be obvious to human doctors. They can help identify patients early that are at risk for heart disease which can help save lives.

2. Data Preparation

There are several important variables in this data set. The variables are as follows: age (the person's age in years), sex (the person's sex: male = 1, female =0), cp (the type of chest pain experienced: no pain = 0, typical angina = 1, atypical angina = 2, non-anginal pain = 3), trestbps (the person's resting blood pressure), chol (the person's cholesterol measurement in mg/dl), fbs (the person's fasting blood sugar is greater than 120 mg/dl: true = 1, false = 0), restecg (resting electrocardiographic measurement: normal = 0, having ST-T wave abnormality = 1, showing probable or definite left ventricular hypertrophy by Estes' criteria), thalach (the person's maximum heart achieved), exang (exercise-induced angina: yes = 1, no = 0), oldpeak (ST depression induced by exercise relative to rest: 'ST' relates to positions on the ECG plot), slope (the slope of the peak exercise ST segment: upsloping = 1, flat = 2, downsloping =3), ca (the number of major vessels: 0 – 3), and target (heart disease: no = 0, yes = 1). There are 303 rows and 14 columns present in this data set.

3. Model #1 - First Logistic Regression Model

Reporting Results

In this section I will write the general form and the prediction equation of the logistic multiple regression model for heart disease using age, resting blood pressure, exercised induced angina, and maximum heart rate. These equations are pasted below. Y is heart disease while x1 is age, x2 is resting blood pressure, x3 is exercise induced angina, and x4 is maximum heart rate.

General Form Equation: $E(y) = \dfrac{e^{(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4)}}{1 + e^{(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4)}}$

Prediction Equation: $E(y) = \dfrac{e^{(-1.0211 - 0.0175 x_1 - 0.0149 x_2 - 1.6250 x_3 + 0.0311 x_4)}}{1 + e^{(-1.0211 - 0.0175 x_1 - 0.0149 x_2 - 1.6250 x_3 + 0.0311 x_4)}}$

From the previous equations I can now write the prediction model equation in terms of the natural log of odds to express the beta terms in linear form. In the equation below the term π is the probability an individual having heart disease, while the term $\frac{\pi}{1-\pi}$ is the ratio of the probability of an individual having heart disease.

$$\ln\left(\frac{\pi}{1-\pi}\right) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4$$

The left side of the equation above is the natural log of odds, so can be written as follows:

Prediction Regression Equation: $\ln(odds) = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \hat{\beta}_3 x_3 + \hat{\beta}_4 x_4$
Linear Form: $\ln(odds) = -1.0211 - 0.0175 x_1 - 0.0149 x_2 - 1.6250 x_3 + 0.0311 x_4$

The estimated coefficient for variable maximum heart rate achieved is 0.0311. This means that on average, the change in log odds for heart disease is 0.000311 for each percentage increases in maximum heart rate achieved, holding all other variables constant. An alternative way to express this in terms of odds is:

$e^{(0.000311)} - 1 = 0.000311$

The odds of heart disease increase by 0.0311% for each percentage increase in maximum heart rate achieved, holding all other variables constant.

Evaluating Model Significance

Using the equation above I performed a Hosmer-Lemeshow goodness of fit test at a 5% level of significance to determine whether this model is appropriate for this data set. The null hypothesis, alternative hypothesis, test statistic, and p-value are pasted below. The p-value is higher than 5% as such we must reject the alternative hypothesis and accept the null hypothesis. I can conclude that this model is appropriate for the data set.

Null Hypothesis: the model is appropriate for the data set, good fit
Alternative Hypothesis: the model is not appropriate for the data set, not good fit
Test Statistic X^2: 44.622
P-Value: 0.612

Following the GOF test I performed a Wald's test at a 5% level of significance to determine which terms are significant in the model. The p-values for the variables exercised-induced angina and maximum heart rate are below 5%. This means that these variables are statistically significant. The variables age and resting blood pressure are above 5% making them not statistically significant.

P-Values-
Age = 0.3060
Trestbps = 0.0741
Exang1 = 1.07e-07
Thalach = 1.92e-05

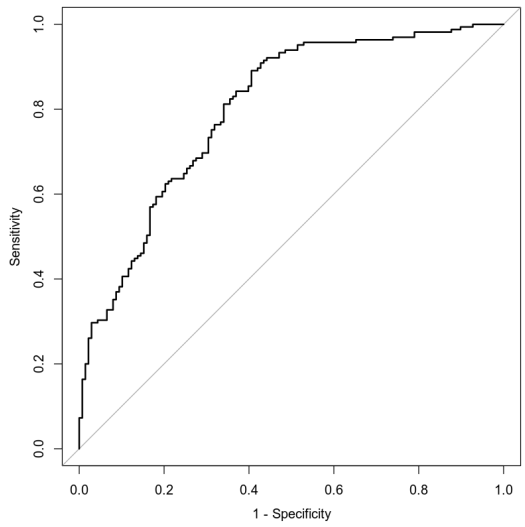
I then went on to obtain the confusion matrix for true positives, true negatives, false positives, and false negatives the results are pasted below.

True Positives = 134
True Negatives = 89
False Positives = 49
False Negatives = 31

The resulting accuracy, precision, and recall resulting from this matrix are pasted below. Accuracy is the ratio of the number of correct predictions to the total number of observations. Precision is the ratio of correct positive predictions to the total predicted positives. Recall is the ratio of correct positive predictions to the total positives examples.

Accuracy = $\frac{TP+TN}{TP+TN+FP+FN} = \frac{134+89}{134+89+49+31} = \frac{223}{303} = 73.60\%$
Precision = $\frac{TP}{TP+FP} = \frac{134}{134+49} = \frac{134}{183} = 73.22\%$
Recall = $\frac{TP}{TP+FN} = \frac{134}{134+31} = \frac{134}{165} = 81.21\%$

The Receiver Operating Characteristic (ROC) curve I obtained is pasted below; the value of the AUC (Area under the Curve) is 0.8007. The ROC curve measures the performance for a classifier at various threshold settings. The area under the curve is an indicator of how well the model distinguishes between y=0 (no heart disease) and y=1 (yes heart disease). In general, a larger area under the curve is desired as it represents a model that is better at predicting the classes. As you can see from the AUC and the graph the model has a good ability to predict whether a patient has heart disease or not.



Making Predictions Using Model

In this section I will be making two predictions using the logistic regression model I created. Recall that a logistic regression model predicts the probability of y = 1. In this model 1 means the patient has heart disease.

Prediction 1: the probability of an individual having heart disease who is 50 years old, has a resting blood pressure of 122, has exercise induced angina, and has maximum heart rate of 140.

The odds of an individual having heart disease in this situation is 0.2716. This indicates that the probability of this patient having heart disease with these indicator variables is 27.16%. This means that this individual is at a low risk for having heart disease.

Prediction 2: the probability of an individual having heart disease who is 50 years old, has a resting blood pressure of 130, does not have an exercise induced angina, and has maximum heart rate of 165.

The odds of an individual having heart disease in this situation is 0.7853. This indicates that the probability of this patient having heart disease with these indicator variables is 78.53%. This means that this individual is at high risk for having heart disease.

4. Model #2 - Second Logistic Regression Model

Reporting Results

In this section I will be creating another logistic regression model. The new general form and the prediction equation for this model are pasted below. In this model heart disease is still the answering variable but we are using age, resting blood pressure, type of chest pain experienced, maximum heart rate achieved, the quadratic term for age, and the interaction term between age and maximum heart rate achieved. The first variable is age represented by x1 and resting heart rate is x2. X3, x4, x5 are

dummy variables for chest pain type. X6 is maximum heart rate achieved, x7 is the quadratic term for age, and x8/x9 is the interaction term between age and maximum heart rate.

General Form Equation: $E(y) = \frac{e^{(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_5 + \beta_6 x_6 + \beta_7 x_7 + \beta_8 x_8 x_9)}}{1 + e^{(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_5 + \beta_6 x_6 + \beta_7 x_7 + \beta_8 x_8 x_9)}}$

Prediction Equation: $E(y)$

$$= \frac{e^{(-15.56 + 0.1744 x_1 - (1.958e-2) 3 x_2 + 1.913 x_3 + 2.037 x_4 + 1.777 x_5 + 0.1363 x_6 + (8.424e-4) x_7 - (1.867e-3) x_8 x_9)}}{1 + e^{(-15.56 + 0.1744 x_1 - (1.958e-2) 3 x_2 + 1.913 x_3 + 2.037 x_4 + 1.777 x_5 + 0.1363 x_6 + (8.424e-4) x_7 - (1.867e-3) x_8 x_9)}}$$

From the previous equations I can now write the prediction model equation in terms of the natural log of odds to express the beta terms in linear form.

$$\ln\left(\frac{\pi}{1-\pi}\right) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_5 + \beta_6 x_6 + \beta_7 x_7 + \beta_8 x_{8x_9}$$

The left side of the equation above is the natural log of odds, so can be written as follows:

Prediction Regression Equation: $\ln(odds) = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \hat{\beta}_3 x_3 + \hat{\beta}_4 x_4 + \hat{\beta}_5 x_5 + \hat{\beta}_6 x_6 + \hat{\beta}_7 x_7 + \hat{\beta}_8 x_8 x_9$

Linear Form: ln(odds) = $-15.56 + 0.1744 x_1 - (1.958e-2) 3 x_2 + 1.913 x_3 + 2.037 x_4 + 1.777 x_5 + 0.1363 x_6 + (8.424e-4) x_7 - (1.867e-3) x_8 x_9$

Evaluating Model Significance

Using the equation above I performed a Hosmer-Lemeshow goodness of fit test at a 5% level of significance to determine whether this model is appropriate for this data set. The null hypothesis, alternative hypothesis, test statistic, and p-value are pasted below. The p-value is higher than 5% as such we must reject the alternative hypothesis and accept the null hypothesis. I can conclude that this model is appropriate for the data set.

Null Hypothesis: the model is appropriate for the data set, good fit
Alternative Hypothesis: the model is not appropriate for the data set, not good fit
Test Statistic X^2: 52
P-Value: 0.3209

Following the GOF test I performed a Wald's test at a 5% level of significance to determine which terms are significant in the model. The p-values for the variables resting blood pressure, typical angina chest pain, atypical angina chest pain, non-anginal chest pain, maximum heart rate, and the interaction term between age and maximum heart rate are below 5%. This means that these variables are statistically significant. The variables age and the quadratic term for age are above 5% making them not statistically significant.

P-Values-
Age = 0.5136
Trestbps = 0.0292
Cp1 = 1.61e-05
Cp2 = 4.45e-09

Cp3 = 0.0012
Thalach = 0.0078
I(age^2) = 0.6303
Age:thalach = 0.0362

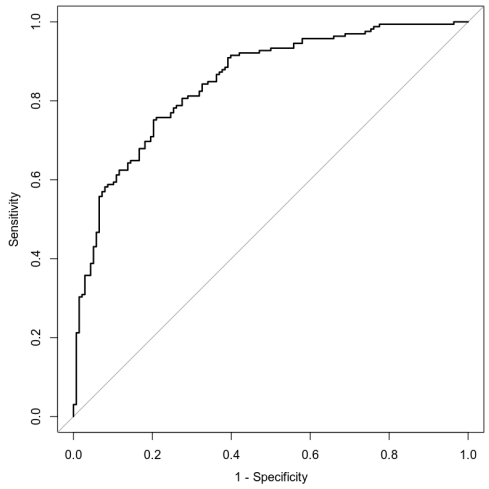
I then went on to obtain the confusion matrix for true positives, true negatives, false positives, and false negatives the results are pasted below.

True Positives: 129
True Negatives: 102
False Positives: 36
False Negatives: 36

The resulting accuracy, precision, and recall resulting from this matrix are pasted below. Accuracy is the ratio of the number of correct predictions to the total number of observations. Precision is the ratio of correct positive predictions to the total predicted positives. Recall is the ratio of correct positive predictions to the total positives examples.

Accuracy = $\frac{TP+TN}{TP+TN+FP+FN} = \frac{129+102}{129+102+36+36} = \frac{231}{303} = 76.24\%$
Precision = $\frac{TP}{TP+FP} = \frac{129}{129+36} = \frac{129}{165} = 78.18\%$
Recall = $\frac{TP}{TP+FN} = \frac{129}{129+36} = \frac{129}{165} = 78.18\%$

The Receiver Operating Characteristic (ROC) curve I obtained is pasted below; the value of the AUC (Area under the Curve) is 0.8478. The ROC curve measures the performance for a classifier at various threshold settings. The area under the curve is an indicator of how well the model distinguishes between y=0 (no heart disease) and y=1 (yes heart disease). In general, a larger area under the curve is desired as it represents a model that is better at predicting the classes. As you can see from the AUC and the graph the model has a good ability to predict whether a patient has heart disease or not.



Making Predictions Using Model

In this section I will be making two predictions using the logistic regression model I created. Recall that a logistic regression model predicts the probability of y = 1. In this model 1 means the patient has heart disease.

Prediction 1: the probability of an individual having heart disease who is 50 years old, has a resting blood pressure of 115, does not experience chest pain, and has maximum heart rate of 133.

The odds of an individual having heart disease in this situation is 0.2188. This indicates that the probability of this patient having heart disease with these indicator variables is 21.88%. This means that this individual is at a low risk for having heart disease.

Prediction 2: the probability of an individual having heart disease who is 50 years old, has a resting blood pressure of 125, experiences typical angina, and has maximum heart rate of 155.

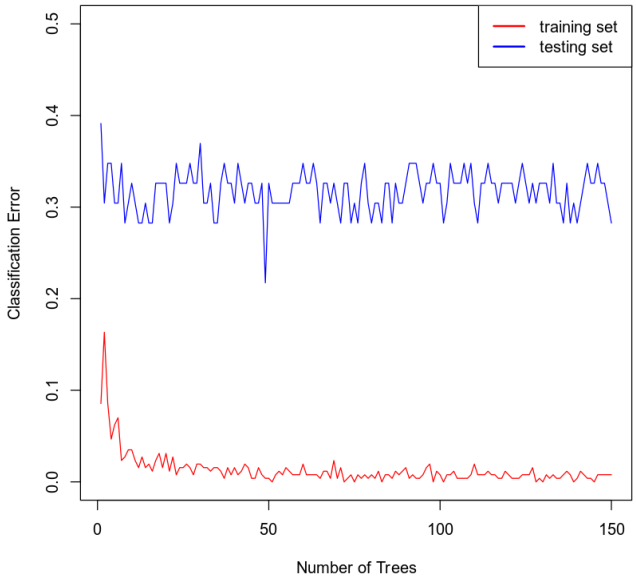
The odds of an individual having heart disease in this situation is 0.8007. This indicates that the probability of this patient having heart disease with these indicator variables is 80.07%. This means that this individual is at a high risk for having heart disease.

5. Random Forest Classification Model

Reporting Results

In this section I will be creating a random forest classification model. In order to do this, I will first split the heart disease data set into training and testing sets using 85% and 15% split respectively. Using a set.seed(6522048) there are 303 rows in the original data set, 257 rows in the training set, and 46 rows for the validation set.

The graph for the training and testing error against the number of trees using a classification random forest model for the presence of heart disease using variables age, sex, chest pain type, resting blood pressure, cholesterol measurement, resting electrocardiographic measurement, exercise-induced angina, and number of major vessels is pasted below. A maximum of 150 trees was used with a set.seed(6522048). From the output, the classification error curve flattens at approximately 25-30 trees. Beyond this, the error is roughly the same even if the number of trees increases. Therefore, I set 30 trees as the optimal number of trees for this random forest classification model.



Evaluating the Utility of the model

Using 30 as the appropriate number trees found in the reporting results section, I created a classification random forest model for the presence of heart disease using variables age, sex, chest pain type, resting blood pressure, cholesterol measurement, resting electrocardiographic measurement, exercise-induced angina, and number of major vessels. I then obtained a confusion matrix for the training set and the testing set. The accuracy, precision, and recall values for each of these sets are pasted below.

Training Set:

True Positives: 136
True Negatives: 119
False Positives: 1
False Negatives: 1

Accuracy = $\frac{TP+TN}{TP+TN+FP+FN} = \frac{136+119}{136+119+1+1} = \frac{255}{257} = 99.22\%$

Precision = $\frac{TP}{TP+FP} = \frac{136}{136+1} = \frac{136}{137} = 99.27\%$

Recall = $\frac{TP}{TP+FN} = \frac{136}{136+1} = \frac{136}{137} = 99.27\%$

Testing Set:

True Positives: 19
True Negatives: 11
False Positives: 7
False Negatives: 9

Accuracy = $\frac{TP+TN}{TP+TN+FP+FN} = \frac{19+11}{19+11+7+9} = \frac{30}{46} = 65.22\%$

Precision = $\frac{TP}{TP+FP} = \frac{19}{19+7} = \frac{19}{26} = 73.08\%$

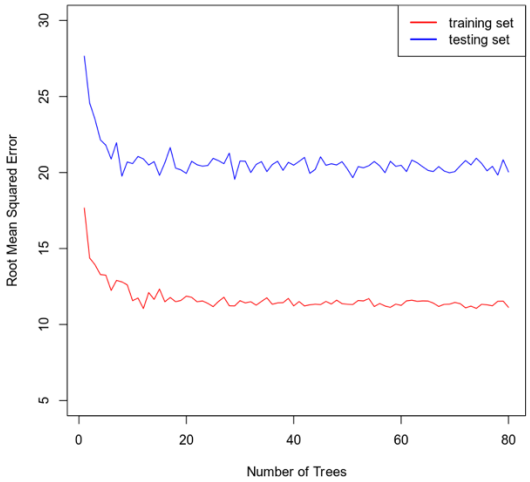
Recall = $\frac{TP}{TP+FN} = \frac{19}{19+9} = \frac{19}{28} = 67.86\%$

6. Random Forest Regression Model

Reporting Results

In this section I will be creating a random forest regression model. In order to do this, I will first split the heart disease data set into training and testing sets using an 80% and 20% split, respectively. Using a set.seed(6522048) there are 303 rows in the original data set, 242 rows in the training set, and 61 rows in the validation set.

The graph for the mean squared error against the number of trees for this model is set to determine maximum heart rate achieved using age, sex, chest pain type, resting blood pressure, cholesterol measurement, resting electrocardiographic measurement, exercise-induced angina, and number of major blood vessels. The graph is pasted below. A maximum of 80 trees was used with a set.seed(6522048). From the output, the root mean squared error curve flattens at approximately 20 trees. Beyond this, the error is roughly the same even if the number of trees increases. Therefore, I set 20 trees as the optimal number of trees for this random forest regression model.



Evaluating the Utility of the Random Forest Regression Model

Using 20 as the appropriate number of trees found, I created a random forest regression model for maximum heart rate achieved using age, sex, chest pain type, resting blood pressure, cholesterol measurement, resting electrocardiographic measurement, exercise-induced angina, and number of major vessels. The root mean squared error for the training set is 11.6282. The root mean squared error for the testing set is 21.1139.

7. Conclusion

I would choose the second logistic regression model to predict heart disease. The second model uses a greater number of variables to analyze an individual's risk of heart disease. Six of these eight terms were proven to be statistically significant while only two of the four variables were statistically significant in the first model. The value of the AUC is also higher for the second model. In general, a larger area under the curve value is desired as it represents a model that is better at predicting the classes.

I would also recommend using the random forest classification model instead of the logistic regression model. Random forest models tend to be more accurate since they can handle more complex

data. You can see that this is the case in this data set too since the accuracy, precision, and recall values are significantly higher for the random forest than the logistic regression models.

These analyses help to improve quality of healthcare for patients. This data can help doctors identify greater risk factors for heart disease. Patients can also be diagnosed earlier and given medication and treatment plans to reduce the danger of heart disease and improve quality of life.